Five-Stage Fourth-Order SDIRK

Skvortsov S54b non-geometric reference method

1 Butcher tableau

For $\dot{z} = f(t, z)$, the S54b tableau is

$$\begin{array}{c|ccccc} \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{8} & \frac{1}{8} & \frac{1}{4} & 0 & 0 \\ 1 & -\frac{3}{2} & \frac{3}{4} & \frac{3}{2} & \frac{1}{4} & 0 \\ 1 & 0 & \frac{1}{6} & \frac{2}{3} & -\frac{1}{12} & \frac{1}{4} \\ \hline & 0 & \frac{1}{6} & \frac{2}{3} & -\frac{1}{12} & \frac{1}{4} \end{array}.$$

All five diagonal entries equal $\gamma = 1/4$, so the method is singly diagonally implicit. The final row of A equals b^T, making the method stiffly accurate.

The stages are solved sequentially:

$$Z_i = z_n + h \sum_{j<i} a_{ij} f(t_n + c_j h, Z_j) + h\gamma f(t_n + c_i h, Z_i), \tag{1}$$

$$z_{n+1} = z_n + h \sum_{i=1}^{5} b_i f(t_n + c_i h, Z_i). \tag{2}$$

Each stage therefore requires a nonlinear solve in the physical state dimension, rather than one simultaneous system containing all stages.

2 Fourth-order conditions

Writing $C = \mathrm{diag}(c)$ and using componentwise powers, the coefficients satisfy

$$\begin{array}{llll} b^T\mathbf{1} = 1, & b^T c = \frac{1}{2}, & b^T c^2 = \frac{1}{3}, & b^T Ac = \frac{1}{6}, \\ b^T c^3 = \frac{1}{4}, & b^T CAc = \frac{1}{8}, & b^T Ac^2 = \frac{1}{12}, & b^T A^2 c = \frac{1}{24}. \end{array}$$

These are the Runge–Kutta conditions through order four. For sufficiently smooth problems the local truncation error is $\mathcal{O}(h^5)$ and the global error is $\mathcal{O}(h^4)$.

3 Deliberately non-geometric structure

A Runge–Kutta method is symplectic for general canonical Hamiltonian systems only if

$$b_i a_{ij} + b_j a_{ji} - b_i b_j = 0 \qquad \text{for every } i, j.$$

The S54b coefficients do not satisfy this identity. For example, with $i = j = 5$ the left-hand side is $1/16$, not zero. Hence S54b is not a symplectic Runge–Kutta method.

Symmetry would require equality with the adjoint tableau, including

$$c_i = 1 - c_{6-i}, \qquad b_i = b_{6-i}, \qquad a_{ij} + a_{6-i,6-j} = b_{6-j}.$$

Already $c_1 = 1/4 \neq 1 - c_5 = 0$, so S54b is not symmetric. These failures make the method a useful same-order control in long-time comparisons against geometric fourth-order methods.

4 Newton implementation

For a known right-hand side r_i, one stage solves

$$R_i(Z_i) = Z_i - r_i - h\gamma f(t_n + c_i h, Z_i) = 0.$$

Newton corrections satisfy

$$\left[I - h\gamma D_z f(t_n + c_i h, Z_i^{(k)})\right] \delta_i^{(k)} = -R_i(Z_i^{(k)}).$$

Execution supplies f and its optional dense or particle-block Jacobian; guiding-centre systems supply independent 2×2 blocks. Configure differentiation with `solver=NewtonSolver( jacobian_method=...)`. Analytic mode uses the supplied derivative, finite-difference mode approximates it from f, and auto uses the supplied Jacobian when available. The solver also owns tolerances and iteration limits. Per-stage and per-step iterations, residual evaluations, residual norms, and effective tolerances are returned as diagnostics.

5 Reference

The coefficient table is the S54b method presented by L. M. Skvortsov, "Diagonally implicit Runge–Kutta methods for stiff problems," *Computational Mathematics and Mathematical Physics*, 46(12), 2110–2123 (2006), DOI: 10.1134/S0965542506120098.

6 Physical state and endpoint energy observations

Accepted states contain one physical vector $Z \in \mathbb{R}^{dN}$ and scalar time. GC vectors have $2N$ entries and FC vectors have $4N$; doubled workspaces are transient. InitialConditions supplies values and layout during preparation only. The numerical base retains functions, metadata, run status and copy count; execution owns the initial snapshot and output schedule. Supplied Jacobians are independent particle blocks (N, d, d); finite differences use dense matrices.

The generic `EnergyObserver` evaluates only

$$H_n = H(t_n, Z_n), \quad H_{n+1} = H(t_{n+1}, Z_{n+1}), \quad (\partial_t H)_n, \quad (\partial_t H)_{n+1}, \quad \Delta H_n = H_{n+1} - H_n.$$

It also evaluates these physical functions at requested saved samples, without consuming internal stages or integrating a momentum coordinate. The derivative is positive, analytic and partial at fixed state. GC differentiates its normalized Hamiltonian potential with `dt=1`; spatial splines and Fourier time factors retain their original scaling. This is not the secant $\Delta H_n/h$. Variation of a time-dependent Hamiltonian is not a numerical conservation error.

Specialized historical generalized-energy studies separately reconstruct $\dot{\kappa} = -\partial_t H$ from retained numerical stages. Their signed weights, physical normalization and archived diagnostics keep their original meaning. They are not part of generic energy observation. Host and JAX execution share the fixed numerical map and physical-only accepted state. JAX evaluates endpoint energy on the device without a momentum accumulator.

7 Optional JAX execution

An optional JAX driver executes the same numerical map on one selected CPU or GPU, with particle operations batched inside a compiled sequential time loop. It reuses the canonical coefficients and prepared potential splines. Physical-state tolerance scaling, signed stage times, passive energy weights and independent shortened output maps retain their definitions above. Newton systems use the supplied particle-block structure. Nonlinear iteration limits and failure checks remain active on the device.

Both execution routes use double precision and return the same physical solution layout. Floating-point reassociation can produce round-off differences without changing the designed order or geometric properties. First-call compilation is separate from warmed integration timings, which include transfers and output construction. The sibling runtime document and shared execution contract specify supported callbacks and exact device scope.